

Checks: 40 pass, 0 warn, 0 fail. All 40 are on the last sheet.

Every check passes.

Sizes

Size	mm
Overall, everything	522.4 × 256.9 × 76.5
bottom, formed	513.8 × 251.9 × 52.2
bottom, flat	542.95 × 340.32
top, formed	518.4 × 256.9 × 57
top, flat	625.55 × 256.9
Between the cheeks	514.4 (keybed 511.4 + 1.5 each side)
White key tops	z = 50.5
Pressed key underside	z = 26.5
Shelf under the keys	z = 21.5
Panel top	z = 54.5

Parts list

#	Component	Qty	Source
1	bottom	1	sheet metal
2	top	1	sheet metal
3	Huaxin 37-key keybed (simplified)	1	STEP: keybed.step
4	screw_M3x6_pan	6	stand-in
5	PEM_CLS_M3	6	stand-in
6	rubber_foot_12x4	4	stand-in
7	pot_9mm	8	stand-in
8	led_3mm	1	stand-in

Parameters

Parameter	Value
shop	jlccnc
material	Aluminum 5052
thickness	2
key_side	1.5
key_front	1.5
key_travel	10
shelf_drop	15
shelf_depth	25
rear_zone	85
key_clear	2
panel_gap	3
recess	2.5
tab	18
fit	0.3
corner	overlap
corner_relief	tear
bend_relief	shear
pots	8

NOTES

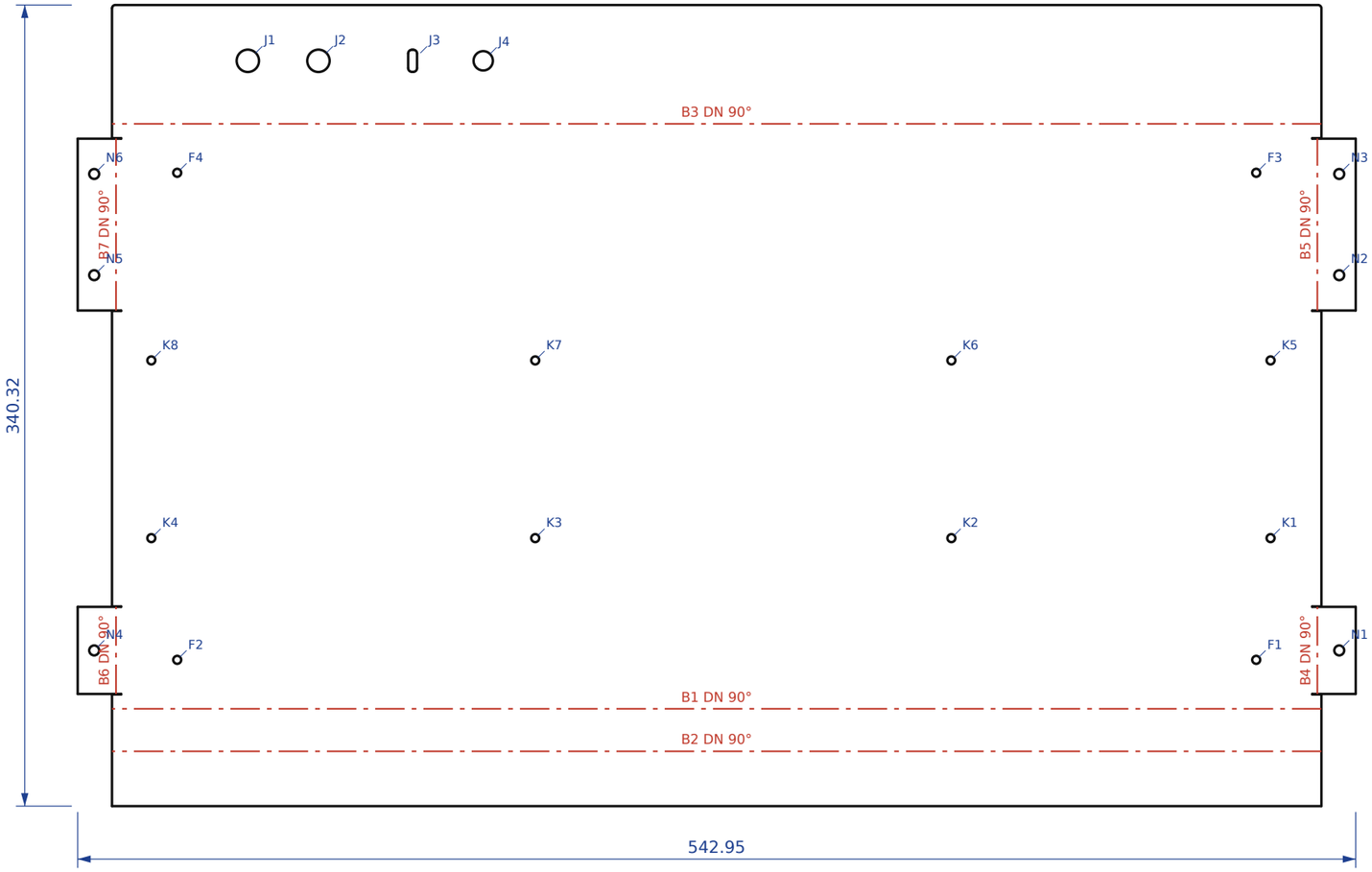
1. Inside bend radius R1.7, K-factor 0.415.
2. Break sharp edges. Deburr all cut edges.
3. Generated by qfold from exact model geometry. Do not scale.
4. Inside radius 1.7 mm = published outer corner radius 3.7 - T.
5. K 0.415 is the middle of JLCCNC's design-guide range, not shop data. JLCCNC unfolds the STEP you send, so K only affects qfold's own flat pattern.
6. Actual sheet thickness 1.95 mm for nominal 2.

Keybed case: assembly

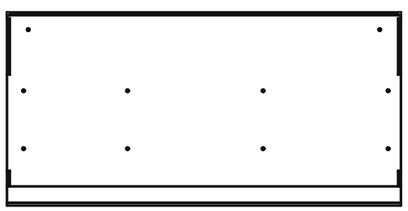
qfold

PART	SCALE	SHEET
ASSEMBLY	NTS	1 of 5
MATERIAL	UNITS	REV / DATE
JLCCNC Aluminum 5052, t=2	mm	A 2026-11-09

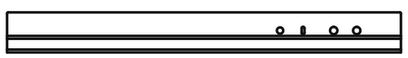
Formed size (outside) 513.8 × 251.9 × 52.2



FLAT PATTERN 542.95 × 340.32



TOP (1:8)



FRONT (1:8)



RIGHT (1:8)

Holes (flat pattern, from lower-left)

Tag	Feature	X	Y	Size
K1	keybed boss screw M3	506.77	113.85	Ø3.4
K2	keybed boss screw M3	371.17	113.85	Ø3.4
K3	keybed boss screw M3	194.37	113.85	Ø3.4
K4	keybed boss screw M3	31.27	113.85	Ø3.4
K5	keybed boss screw M3	506.77	189.35	Ø3.4
K6	keybed boss screw M3	371.17	189.35	Ø3.4
K7	keybed boss screw M3	194.37	189.35	Ø3.4
K8	keybed boss screw M3	31.27	189.35	Ø3.4
F1	foot screw M3	500.67	62.15	Ø3.4
F2	foot screw M3	42.27	62.15	Ø3.4
F3	foot screw M3	500.67	269.05	Ø3.4
F4	foot screw M3	42.27	269.05	Ø3.4
J1	1/4" jack	72.27	316.62	Ø9.5
J2	1/4" jack	102.27	316.62	Ø9.5
J3	USB-C	142.27	316.62	9.4×3.8 R1.6
J4	DC barrel jack	172.27	316.62	Ø8.2
N1	PEM CLS-M3 nut	535.95	66.15	Ø4.22
N2	PEM CLS-M3 nut	535.95	225.55	Ø4.22
N3	PEM CLS-M3 nut	535.95	268.55	Ø4.22
N4	PEM CLS-M3 nut	7	66.15	Ø4.22
N5	PEM CLS-M3 nut	7	225.55	Ø4.22
N6	PEM CLS-M3 nut	7	268.55	Ø4.22

Bends

Tag	Bend	Dir	Angle	In. R	Length	Allow.	Centre line
B1	front_wall	DN	90°	R1.7	513.8	3.97	Y 41.36
B2	key_shelf	DN	90°	R1.7	513.8	3.97	Y 23.29
B3	back_wall	DN	90°	R1.7	513.8	3.97	Y 289.84
B4	left_tab_front	DN	90°	R1.7	37	3.97	X 526.66
B5	left_tab_rear	DN	90°	R1.7	73	3.97	X 526.66
B6	right_tab_front	DN	90°	R1.7	37	3.97	X 16.29
B7	right_tab_rear	DN	90°	R1.7	73	3.97	X 16.29

NOTES

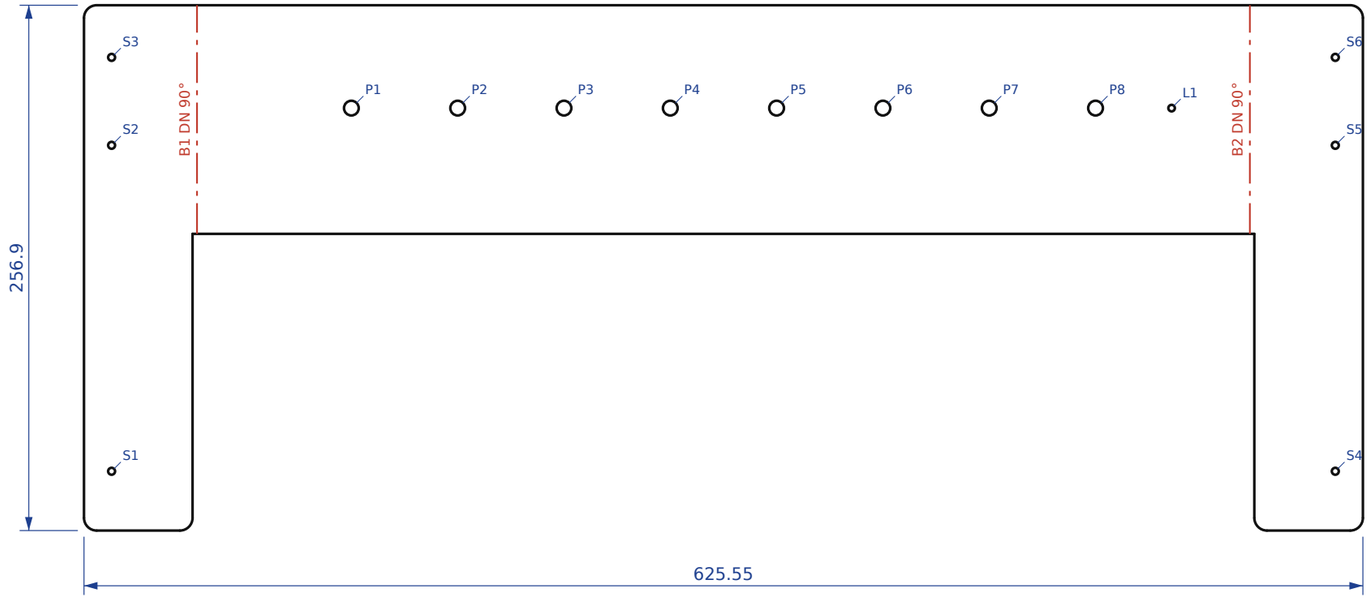
1. Inside bend radius R1.7, K-factor 0.415.
2. Break sharp edges. Deburr all cut edges.
3. Generated by qfold from exact model geometry. Do not scale.
4. Inside radius 1.7 mm = published outer corner radius 3.7 - T.
5. K 0.415 is the middle of JLCCNC's design-guide range, not shop data. JLCCNC unfolds the STEP you send, so K only affects qfold's own flat pattern.
6. Actual sheet thickness 1.95 mm for nominal 2.
7. Flat pattern: outside face shown. DN = bend away from viewer. The DXF is the cutting master; the STEP is the formed part.

bottom

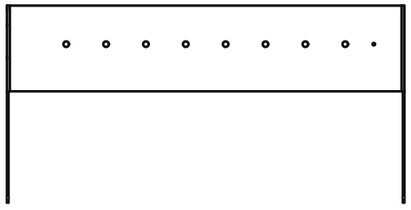
qfold

PART	SCALE	SHEET
BOTTOM	1:2.5	2 of 5
MATERIAL	UNITS	REV / DATE
JLCCNC Aluminum 5052, t=2	mm	A 2026-11-09

Formed size (outside) 518.4 × 256.9 × 57



FLAT PATTERN 625.55 × 256.9



TOP (1:8)



FRONT (1:8)



RIGHT (1:8)

Holes (flat pattern, from lower-left)

Tag	Feature	X	Y	Size
P1	pot (M7 bushing)	130.77	206.59	Ø7.2
P2	pot (M7 bushing)	182.77	206.59	Ø7.2
P3	pot (M7 bushing)	234.77	206.59	Ø7.2
P4	pot (M7 bushing)	286.77	206.59	Ø7.2
P5	pot (M7 bushing)	338.77	206.59	Ø7.2
P6	pot (M7 bushing)	390.77	206.59	Ø7.2
P7	pot (M7 bushing)	442.77	206.59	Ø7.2
P8	pot (M7 bushing)	494.77	206.59	Ø7.2
L1	3 mm LED	531.97	206.59	Ø3.2
S1	M3 clearance	13.5	29	Ø3.4
S2	M3 clearance	13.5	188.4	Ø3.4
S3	M3 clearance	13.5	231.4	Ø3.4
S4	M3 clearance	612.05	29	Ø3.4
S5	M3 clearance	612.05	188.4	Ø3.4
S6	M3 clearance	612.05	231.4	Ø3.4

Bends

Tag	Bend	Dir	Angle	In. R	Length	Allow.	Centre line
B1	left_cheek	DN	90°	R1.7	111.8	3.97	X 55.29
B2	right_cheek	DN	90°	R1.7	111.8	3.97	X 570.26

NOTES

1. Inside bend radius R1.7, K-factor 0.415.
2. Break sharp edges. Deburr all cut edges.
3. Generated by qfold from exact model geometry. Do not scale.
4. Inside radius 1.7 mm = published outer corner radius 3.7 - T.
5. K 0.415 is the middle of JLCCNC's design-guide range, not shop data. JLCCNC unfolds the STEP you send, so K only affects qfold's own flat pattern.
6. Actual sheet thickness 1.95 mm for nominal 2.
7. Flat pattern: outside face shown. DN = bend away from viewer. The DXF is the cutting master; the STEP is the formed part.

top		qfold
PART **TOP**	SCALE 1:3	SHEET 3 of 5
MATERIAL JLCCNC Aluminum 5052, t=2	UNITS mm	REV / DATE A 2026-11-09

Part	Tag	Feature	Size	X	Y	Z	Face	Facing
bottom	K1	keybed boss screw M3	Ø3.4	23.9	76.7	0	base	−Z down
bottom	K2	keybed boss screw M3	Ø3.4	159.5	76.7	0	base	−Z down
bottom	K3	keybed boss screw M3	Ø3.4	336.3	76.7	0	base	−Z down
bottom	K4	keybed boss screw M3	Ø3.4	499.4	76.7	0	base	−Z down
bottom	K5	keybed boss screw M3	Ø3.4	23.9	152.2	0	base	−Z down
bottom	K6	keybed boss screw M3	Ø3.4	159.5	152.2	0	base	−Z down
bottom	K7	keybed boss screw M3	Ø3.4	336.3	152.2	0	base	−Z down
bottom	K8	keybed boss screw M3	Ø3.4	499.4	152.2	0	base	−Z down
bottom	F1	foot screw M3	Ø3.4	30	25	0	base	−Z down
bottom	F2	foot screw M3	Ø3.4	488.4	25	0	base	−Z down
bottom	F3	foot screw M3	Ø3.4	30	231.9	0	base	−Z down
bottom	F4	foot screw M3	Ø3.4	488.4	231.9	0	base	−Z down
bottom	J1	1/4" jack	Ø9.5	458.4	254.4	28.5	back_wall	+Y back
bottom	J2	1/4" jack	Ø9.5	428.4	254.4	28.5	back_wall	+Y back
bottom	J3	USB-C	9.4×3.8 R1.6	388.4	254.4	28.5	back_wall	+Y back
bottom	J4	DC barrel jack	Ø8.2	358.4	254.4	28.5	back_wall	+Y back
bottom	N1	PEM CLS-M3 nut	Ø4.22	2.3	29	11	left_tab_front	−X left
bottom	N2	PEM CLS-M3 nut	Ø4.22	2.3	188.4	11	left_tab_rear	−X left
bottom	N3	PEM CLS-M3 nut	Ø4.22	2.3	231.4	11	left_tab_rear	−X left
bottom	N4	PEM CLS-M3 nut	Ø4.22	516.1	29	11	right_tab_front	+X right
bottom	N5	PEM CLS-M3 nut	Ø4.22	516.1	188.4	11	right_tab_rear	+X right
bottom	N6	PEM CLS-M3 nut	Ø4.22	516.1	231.4	11	right_tab_rear	+X right
top	P1	pot (M7 bushing)	Ø7.2	77.2	206.59	54.5	panel	+Z up
top	P2	pot (M7 bushing)	Ø7.2	129.2	206.59	54.5	panel	+Z up
top	P3	pot (M7 bushing)	Ø7.2	181.2	206.59	54.5	panel	+Z up
top	P4	pot (M7 bushing)	Ø7.2	233.2	206.59	54.5	panel	+Z up
top	P5	pot (M7 bushing)	Ø7.2	285.2	206.59	54.5	panel	+Z up
top	P6	pot (M7 bushing)	Ø7.2	337.2	206.59	54.5	panel	+Z up
top	P7	pot (M7 bushing)	Ø7.2	389.2	206.59	54.5	panel	+Z up
top	P8	pot (M7 bushing)	Ø7.2	441.2	206.59	54.5	panel	+Z up
top	L1	3 mm LED	Ø3.2	478.4	206.59	54.5	panel	+Z up
top	S1	M3 clearance	Ø3.4	0	29	11	left_cheek	−X left
top	S2	M3 clearance	Ø3.4	0	188.4	11	left_cheek	−X left
top	S3	M3 clearance	Ø3.4	0	231.4	11	left_cheek	−X left
top	S4	M3 clearance	Ø3.4	518.4	29	11	right_cheek	+X right
top	S5	M3 clearance	Ø3.4	518.4	188.4	11	right_cheek	+X right
top	S6	M3 clearance	Ø3.4	518.4	231.4	11	right_cheek	+X right

NOTES

1. Hole centres on the outside surface, in assembly coordinates.
2. Facing: the direction the hole opens to, out of the part.

Holes as assembled

qfold

PART	SCALE	SHEET
KEYBED CASE	NTS	4 of 5
MATERIAL	UNITS	REV / DATE
JLCCNC Aluminum 5052, t=2	mm	A 2026-11-09

Checks: 40 pass, 0 warn, 0 fail

Check	Result	
bottom: valid solid	valid=True, solids=1, faces=97	PASS
bottom: no self-intersection	folded fuse loses -5.82e-11 mm³, flat fuse loses -1.16e-10 mm³ (limit 3.6e-04)	PASS
bottom: flat pattern matches fold	folded−flat volume 940.712221 mm³, bend allowance predicts 940.712221	PASS
bottom: hole to bend	tightest 5.19 mm (left_tab_front PEM CLS-M3 nut to bend left_tab_front), need ≥ 4.00	PASS
bottom: hole to edge	closest 4.89 mm (left_tab_front PEM CLS-M3 nut), need ≥ 3.00	PASS
bottom: web between holes	narrowest 20.50 mm (back_wall 1/4" jack / 1/4" jack), need ≥ 1.00	PASS
bottom: holes inside their face	every hole lies within its face	PASS
bottom: minimum hole size	all holes ≥ 1.00 mm	PASS
bottom: hardware edge distance	6 hardware holes meet their centre-to-edge minimum	PASS
bottom: bend corner relief	bends end ≥ 4.30 mm apart, need ≥ 2.00	PASS
bottom: flange length	shortest flat flange 14.10 mm (bend front_wall), need ≥ 6.30	PASS
top: valid solid	valid=True, solids=1, faces=39	PASS
top: no self-intersection	folded fuse loses 2.91e-10 mm³, flat fuse loses 8.73e-11 mm³ (limit 1.7e-04)	PASS
top: flat pattern matches fold	folded−flat volume 119.418220 mm³, bend allowance predicts 119.418220	PASS
top: hole to bend	tightest 34.70 mm (panel 3 mm LED to bend right_cheek), need ≥ 4.00	PASS
top: hole to edge	closest 11.80 mm (left_cheek M3 clearance), need ≥ 3.00	PASS
top: web between holes	narrowest 32.00 mm (panel pot (M7 bushing) / 3 mm LED), need ≥ 1.00	PASS
top: holes inside their face	every hole lies within its face	PASS
top: minimum hole size	all holes ≥ 1.00 mm	PASS
top: bend corner relief	bends end ≥ 511.00 mm apart, need ≥ 2.00	PASS
top: flange length	shortest flat flange 53.27 mm (bend left_cheek), need ≥ 6.30	PASS
bottom.step reads back	1 solid(s), volume 356212.466 mm³ (model 356212.466)	PASS
top.step reads back	1 solid(s), volume 169935.775 mm³ (model 169935.775)	PASS
assembly.step reads back	74 solid(s), volume 2454406.839 mm³ (model 2454406.839)	PASS
keybed stands on the floor	1 touching, widest gap 0.0000 mm (≤ 0.01), no overlap	PASS
keys and frame clear the top and the nuts	overlap 0.000000 mm³, closest 1.500 mm, need ≥ 1	PASS
pressed keys clear the case	overlap 0.000000 mm³, closest 1.500 mm, need ≥ 1	PASS
top slides over the bottom	overlap 0.000000 mm³, closest 0.300 mm, need ≥ 0.2	PASS
PEM nuts in the tabs	6 touching, widest gap 0.0000 mm (≤ 0.01), no overlap	PASS
screw heads on the cheeks	6 touching, widest gap 0.0000 mm (≤ 0.01), no overlap	PASS
feet on the floor	4 touching, widest gap 0.0000 mm (≤ 0.01), no overlap	PASS
pots against the panel	8 touching, widest gap 0.0000 mm (≤ 0.01), no overlap	PASS
LED in its hole	1 clear of it, no overlap	PASS
faces of one part keep their gaps	22 unjoined face pairs, tightest 1.500 mm (bottom: key_shelf vs left_tab_front), need ≥ 0.2	PASS
no part overlaps another	29 top-level parts, none overlapping	PASS
screws through the cheeks into their nuts	6 parts, worst 0.000 mm across, 0.000 mm along, 0.000°	PASS
bottom: fits sheet	flat 542.9 × 340.3 mm, sheet limit 800 × 800	PASS
bottom_flat.dxf reads back	21 circles, 40 lines, 6 arcs, 7 bends, cut 2202.9 mm, extent 542.948 × 340.322	PASS
top: fits sheet	flat 625.5 × 256.9 mm, sheet limit 800 × 800	PASS
top_flat.dxf reads back	15 circles, 8 lines, 6 arcs, 2 bends, cut 2294.2 mm, extent 625.548 × 256.900	PASS

NOTES

1. Run on every build; also in report.json.
2. Limits come from the project's rules (a shop's, or qfold's defaults in dfm.py).

Checks		qfold
PART **KEYBED CASE**	SCALE NTS	SHEET 5 of 5
MATERIAL JLCCNC Aluminum 5052, t=2	UNITS mm	REV / DATE A 2026-11-09